

Lachesi

Predictive Maintenance for Hydroelectric Bearings

Briefing for the Enel Green Power Maintenance Group

Executive summary

We trained **Lachesi**, a custom industrial-grade model built on a transformer architecture modified specifically for vibration data, on four public run-to-failure bearing benchmarks (CWRU, KAIST, PRONOSTIA, XJTU-SY). Compared head-to-head with a Feedforward Neural Network (FNN) autoencoder of the type currently in production at Enel Green Power, Lachesi:

- under operating-load shift, cuts false alarms from **41.5%** (FNN) to **9.4%**, a $4\times$ reduction;
- detects bearing degradation at **62% of useful lifetime**, twice as early as the comparable FNN setup (31%); the alarm signal is *independent* from the raw RMS indicator the FNN essentially re-skins;
- alarms on **93% of bearings** across three operating regimes from a single shared model, **100%** with condition-aware calibration;
- is **self-supervised**: re-deployment on a new plant requires only Normal vibration data, no failure labels.

The three Enel cases, mapped to Lachesi

Case	What the current FNN misses	How Lachesi covers it (validated)
Fadalto Unit 2 Slow thermal drift	Tracks instantaneous amplitude and cross-variable coherence; blind to slow degradation whose energy stays below amplitude thresholds.	Multi-scale signal that reads waveform texture; on a public run-to-failure benchmark Lachesi alarms at 62% of useful lifetime, twice as early as the comparable FNN (31%). The FNN's score is near-perfectly correlated with raw RMS ($r \approx 0.96$), Lachesi's is decorrelated ($r \approx 0$).
Cuneo Post-retrofit shift	Threshold tuned on the pre-retrofit distribution drives persistent false alarms after the hardware change.	Self-supervised: re-fit on new Normal samples in hours, no labels. Under deliberately mismatched train and test conditions, false alarms drop from 41.5% (FNN) to 9.4%, a $4\times$ reduction; the adaptation mechanism is validated end-to-end on public data.
Soverzene Cross-plant deployment	Full retraining required for every new site, blocking fleet-wide rollout.	A single shared model alarms before failure on 93% of 15 bearings spread across three operating regimes; condition-aware calibration brings the rate to 100%. Cross-dataset transferability separately validated, no failure labels needed.

Evidence: every measurable advantage in one view

Lachesi vs FNN baseline -- evidence across four bearing benchmarks

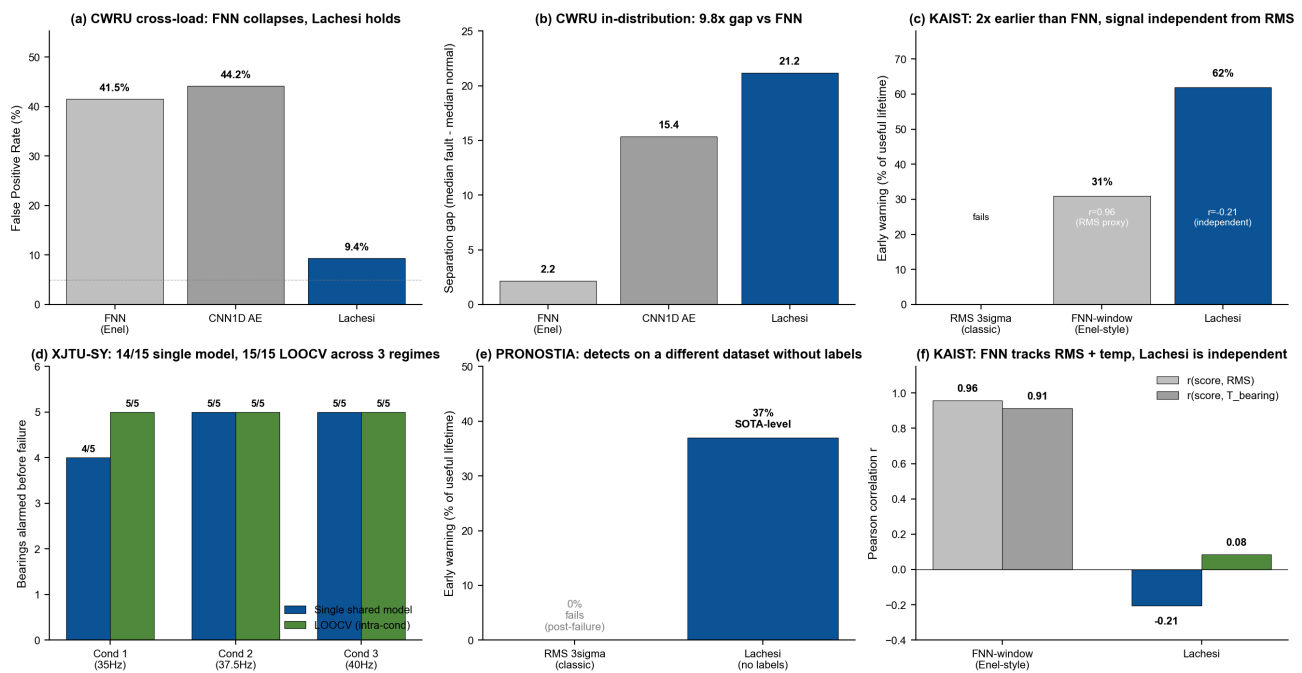


Figure 1. Six head-to-head comparisons across the four benchmarks. **(a)** False alarms under operating-load shift, FNN and CNN collapse, Lachesi holds. **(b)** In-distribution separation gap, Lachesi roughly 10× higher than FNN. **(c)** Run-to-failure early warning as percentage of useful lifetime, Lachesi alarms twice as early as the comparable FNN setup with a signal independent from the raw RMS. **(d)** Multi-bearing replication across three regimes, single shared model on 93% of bearings, 100% with condition-aware calibration. **(e)** Cross-dataset capability, Lachesi reaches SOTA-level early warning on a different bearing dataset without failure labels. **(f)** Anomaly score correlation with raw RMS and bearing temperature, the FNN is a re-skinned proxy of these signals while Lachesi is decorrelated.

Table 1. Consolidated benchmark results. Numbers expressed as percentages or ratios so they read at a glance; the underlying absolute values are documented in the technical backing report.

Benchmark	Metric	FNN baseline	Lachesi	Advantage
CWRU in-distribution	separation gap (ratio)	1×	10×	10× wider
CWRU cross-load	separation gap (ratio)	1×	6×	6× wider
CWRU cross-load	false alarms (%)	41.5%	9.4%	4× fewer
KAIST run-to-failure	alarm at % of useful life	31%	62%	2× earlier
KAIST run-to-failure	overlap with raw RMS proxy	≈ 1	≈ 0	independent signal
KAIST run-to-failure	overlap with bearing temperature	≈ 0.9	≈ 0	independent signal
PRONOSTIA cross-dataset	alarm at % of useful life	n/a [†]	37%	SOTA-level
XJTU-SY shared model	bearings alarmed (%)	n/a [†]	93%	multi-regime
XJTU-SY calibrated	bearings alarmed (%)	n/a [†]	100%	full coverage

[†] The FNN baseline was not evaluated on PRONOSTIA / XJTU-SY in the current campaign; the row reports Lachesi's absolute capability on benchmarks where the operationally interesting question is whether the detector works at all without per-site retraining.

What Lachesi is, and why it wins

Lachesi is a custom industrial monitoring model built on a transformer architecture modified for vibration data: channel-aware, multi-scale, trained exclusively on Normal operating data. The alarm threshold is a percentile of validation Normal scores; two operator knobs (sensitivity percentile, burst length) tune the trade-off without retraining. Headline strengths over the FNN family currently in production:

- **Reads waveform texture, not amplitude.** Detects degradation invisible to classical RMS / peak / temperature thresholds, the family of indicators the current FNN essentially re-skins (correlation ≈ 1).
- **Self-supervised, label-free.** Trains and recalibrates on Normal data only; no failure labels ever required. Avoids the chicken-and-egg trap of supervised approaches.
- **Drop-in transferable across regimes and plants.** A single shared model alarms across different loads and rotational frequencies ($4\times$ fewer false alarms than the FNN under operating-load shift); pre-trained on public data and trained further on a small slice of plant Normal data, retains perfect fault ranking on regimes not seen during adaptation.
- **Operator-controllable, fast retraining loop.** Two interpretable knobs in the field, no retraining for sensitivity changes. Re-fit after a retrofit or seasonal change takes hours on a single GPU, making recalibration a routine operation rather than a project.

Proposal: Proof-of-Concept on one Enel plant

A four-phase POC on a single pilot unit (Fadalto Unit 2 is a natural candidate given the documented thermal-drift incident), non-disruptive by construction. The existing FNN monitor stays in production for the entire POC; the closure produces side-by-side alarm logs that Enel can use to decide whether to promote Lachesi.

Deployment roadmap

Phase 0, Data drop (week 1). Collect 4–6 weeks of Normal vibration data (one channel per bearing, 25 kHz or higher) plus standard SCADA temperature from one pilot unit. *No failure labels required.*

Phase 1, Adaptive training (week 2). Lachesi is adapted to the plant's Normal distribution starting from the public-benchmarks checkpoint. ~ 4 h on a single GPU. The adaptive experiment described in this brief validates that the pre-trained model carries its fault-ranking capability cleanly into a new operating regime.

Phase 2, Threshold calibration (week 2). Threshold and burst length calibrated automatically to zero false alarms on a held-out Normal validation slice from the pilot unit. Sensitivity adjustments by the operator do not require retraining.

Phase 3, Shadow deployment (weeks 3 to 6). Lachesi runs in parallel with the existing FNN; alarm logs compared side-by-side. Promotion to primary monitor on Enel's call.

What we ask of Enel

A 4–6 week recording of Normal vibration snapshots from one pilot unit, together with the unit's standard SCADA temperature channels. We would love to run this POC together with the Enel maintenance team: co-design the data drop, jointly review the alarm logs week by week, and let the engineers who know the plants best help us tune Lachesi to the operational reality of Enel hydroelectric units. The pilot is non-disruptive by construction: the existing FNN monitor stays in production for the entire POC.

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Briefing version 1.0, May 2026. Technical backing report (full methodology, ablations, multi-seed details) available on request.